

Airline Satisfaction Analysis

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Load Libraries

```
library(MASS) library(psych)
```

```
## Warning: package 'psych' was built under R version 4.4.3
```

```
library(corrplot)
```

```
## Warning: package 'corrplot' was built under R version 4.4.3
```

```
## corrplot 0.95 loaded
```

```
library(caret)
```

```
## Warning: package 'caret' was built under R version 4.4.3
```

```
## Loading required package: ggplot2
```

```
##
```

```
## Attaching package: 'ggplot2'
```

```
## The following objects are masked from 'package:psych':
```

```
##
```

```
##      %+%, alpha
```

```
## Loading required package: lattice
```

```
library(pROC)
```

```
## Warning: package 'pROC' was built under R version 4.4.3 ## Type
```

```
'citation("pROC")' for a citation.
```

```
##
```

```
## Attaching package: 'pROC'
```

```
## The following objects are masked from 'package:stats':
```

```
##
```

```
##          cov, smooth, var
```

```
library(ggplot2)
```

Load Dataset

```
train <- read.csv("E:/Downloads/airplane_train/airplane_train.csv") test <-  
read.csv("E:/Downloads/airplane_test/airplane_test.csv")
```

Data Preprocessing

This step removes unnecessary columns, converts categorical variables into factors, aligns factor levels between train and test datasets, and removes missing values. These operations ensure the data is clean and ready for modeling.

```
train <- train[, !names(train) %in% c("X", "id")] test <- test[,  
!names(test) %in% c("X", "id")]
```

```
cat_cols <- c("Gender",  
             "Customer.Type",  
             "Type.of.Travel",  
             "Class",  
             "satisfaction")
```

```
for(col in cat_cols){ train[[col]] <-  
  as.factor(train[[col]]) test[[col]] <-  
  as.factor(test[[col]])  
}
```

```
for(col in cat_cols){ test[[col]] <- factor(test[[col]], levels =  
  levels(train[[col]]))  
}
```

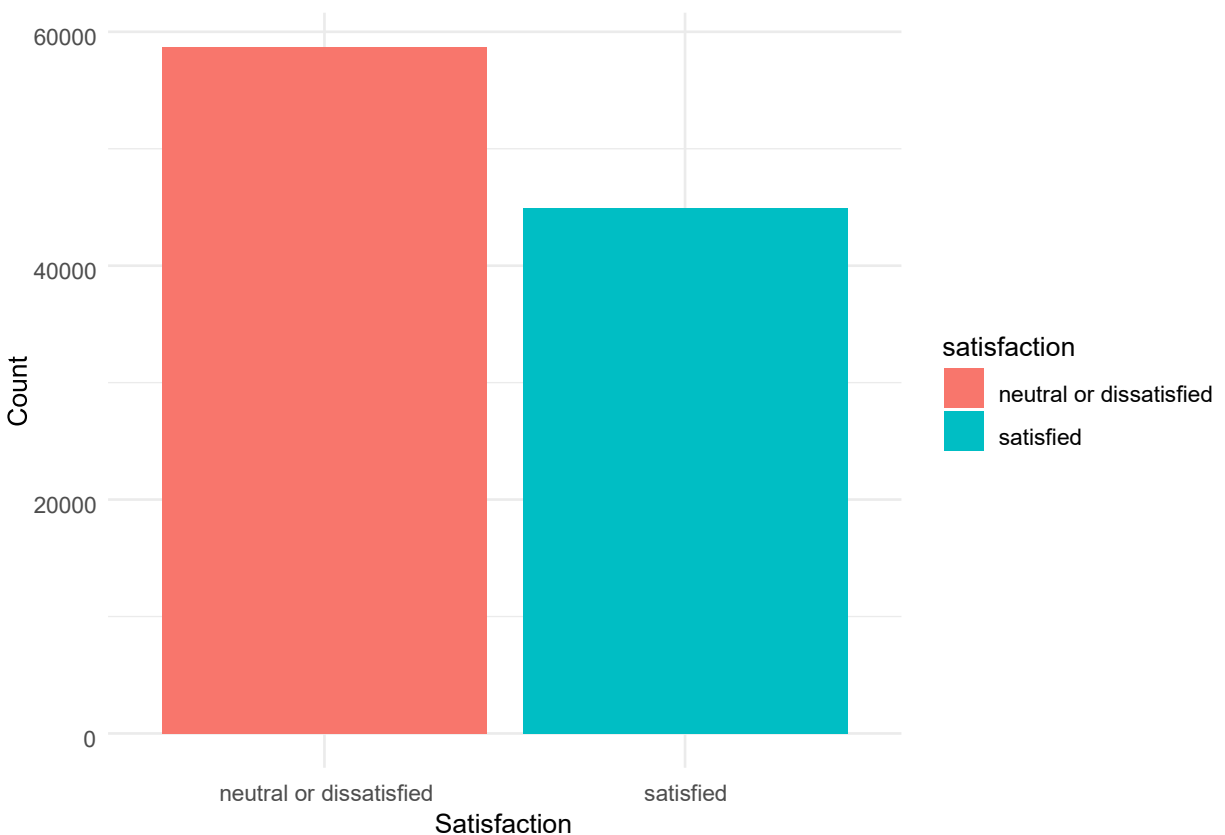
```
train <- na.omit(train) test <-  
na.omit(test)
```

Passenger Satisfaction Distribution

The bar chart shows the distribution of satisfied and dissatisfied passengers in the dataset. The results indicate that there are slightly more neutral or dissatisfied passengers than satisfied passengers.

```
ggplot(train, aes(x=satisfaction,  
                 fill=satisfaction)) +  
  geom_bar() + theme_minimal() + labs( title="Distribution of  
Passenger Satisfaction", x="Satisfaction", y="Count"  
)
```

Distribution of Passenger Satisfaction



Logistic Regression Model

This code builds a Logistic Regression model to predict passenger satisfaction using all available passenger and service-related variables. The model helps identify factors that influence customer satisfaction.

```
log_model <- glm(
  satisfaction ~ ., data =
  train, family = binomial
) summary(log_model)
```

The model summary shows the effect of each variable on passenger satisfaction. Factors such as customer loyalty, online boarding, cleanliness, and inflight services positively influence satisfaction, while personal travel and lower travel classes negatively affect satisfaction. Most variables are statistically significant.

Call:

```
## glm(formula = satisfaction ~ ., family = binomial, data = train)
```

##

Coefficients:

##

(Intercept)

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-7.860e+00	7.876e-02	-99.793	< 2e-16 ***

GenderMale

GenderMale	4.255e-02	1.949e-02	2.183	0.02905 *
------------	-----------	-----------	-------	-----------

Customer.TypeLoyal Customer

Customer.TypeLoyal	2.035e+00	2.994e-02	67.970	< 2e-16 ***
--------------------	-----------	-----------	--------	-------------

Age

Age	-8.308e-03	7.110e-04	-11.684	< 2e-16 ***
-----	------------	-----------	---------	-------------

Type.of.TravelPersonal Travel

Type.of.TravelPersonal	-2.722e+00	3.147e-02	-86.494	< 2e-16 ***
------------------------	------------	-----------	---------	-------------

```

## ClassEco -7.389e-01 2.566e-02 -28.794 < 2e-16 ***
## ClassEco Plus -8.554e-01 4.155e-02 -20.588 < 2e-16 ***
## Flight.Distance -1.789e-05 1.132e-05 -1.581 0.11392
## Inflight.wifi.service 3.949e-01 1.148e-02 34.405 < 2e-16 ***
## Departure.Arrival.time.convenient -1.244e-01 8.218e-03 -15.132 < 2e-16 ***
## Ease.of.Online.booking -1.440e-01 1.135e-02 -12.691 < 2e-16 ***
## Gate.location 2.914e-02 9.174e-03 3.176 0.00149 **
## Food.and.drink -2.860e-02 1.068e-02 -2.677 0.00743 **
## Online.boarding 6.126e-01 1.025e-02 59.773 < 2e-16 ***
## Seat.comfort 6.555e-02 1.118e-02 5.862 4.58e-09 ***
## Inflight.entertainment 6.555e-02 1.427e-02 4.594 4.34e-06 ***
## On.board.service 3.014e-01 1.019e-02 29.582 < 2e-16 ***
## Leg.room.service 2.532e-01 8.540e-03 29.652 < 2e-16 ***
## Baggage.handling 1.331e-01 1.144e-02 11.633 < 2e-16 ***
## Checkin.service 3.234e-01 8.566e-03 37.757 < 2e-16 ***
## Inflight.service 1.207e-01 1.205e-02 10.018 < 2e-16 ***
## Cleanliness 2.236e-01 1.210e-02 18.471 < 2e-16 ***
## Departure.Delay.in.Minutes 4.759e-03 9.882e-04 4.815 1.47e-06 ***
## Arrival.Delay.in.Minutes -9.412e-03 9.745e-04 -9.659 < 2e-16 ***
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1) ##
## Null deviance: 141768 on 103593 degrees of freedom
## Residual deviance: 69169 on 103570 degrees of freedom
## AIC: 69217
##
## Number of Fisher Scoring iterations: 6

```

Confusion Matrix

The confusion matrix evaluates the model's prediction performance. The Logistic Regression model achieved an accuracy of 87.17%, demonstrating strong capability in classifying passenger satisfaction correctly.

```

prob <- predict(log_model,
                test, type="response")

pred <- ifelse( prob >
  0.5,
  "satisfied",
  "neutral or dissatisfied"
)

log_pred <- factor( pred, levels =
  levels(test$satisfaction)
)

cm <- confusionMatrix( log_pred,
  test$satisfaction
)
cm

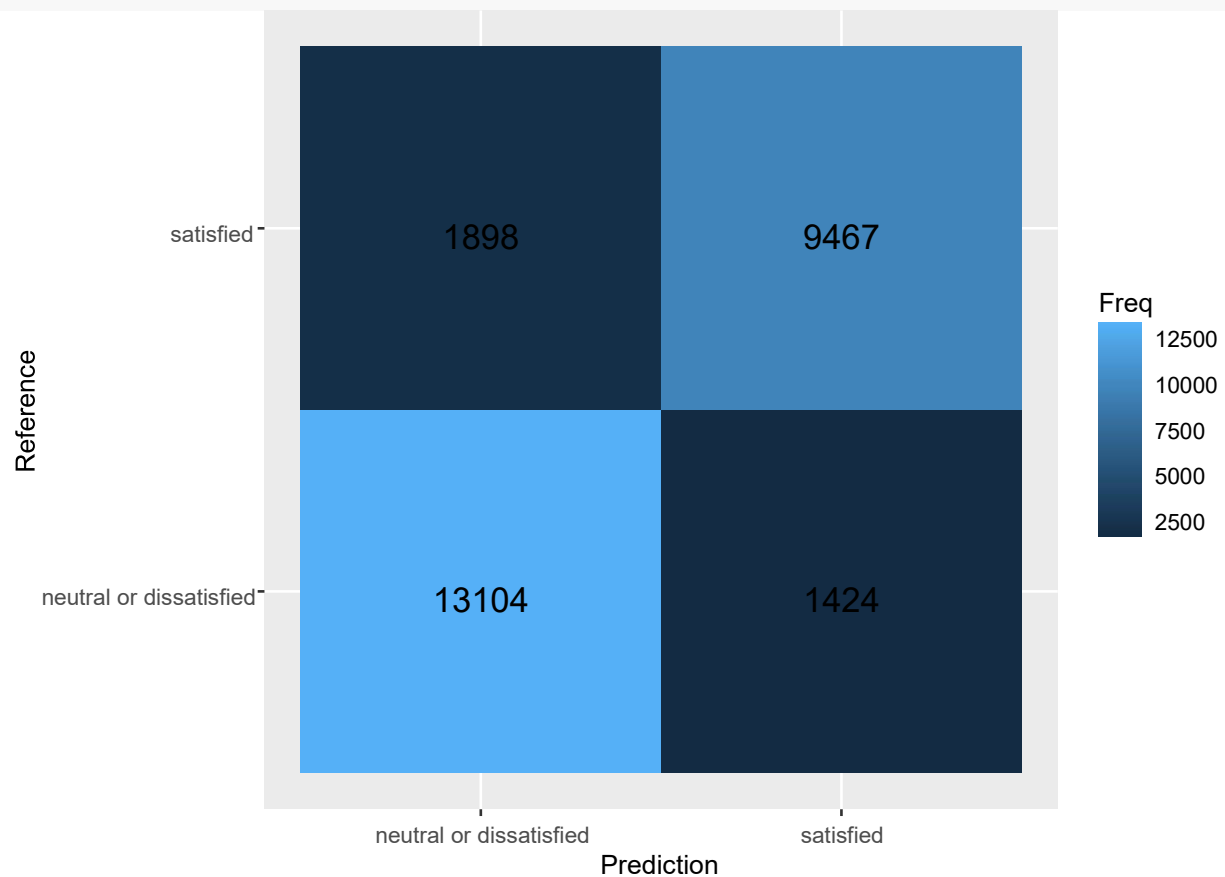
## Confusion Matrix and Statistics
##
##                                Reference
## Prediction                    neutral or dissatisfied satisfied
##      neutral or dissatisfied                    13104      1898
##      satisfied                                1424      9467
##
##              Accuracy : 0.8717
##              95% CI : (0.8676, 0.8758)
##      No Information Rate : 0.5611
##      P-Value [Acc > NIR] : < 2.2e-16
##
##              Kappa : 0.7383
##
## Mcnemar's Test P-Value : 2.276e-16
##
##              Sensitivity : 0.9020
##              Specificity : 0.8330
##              Pos Pred Value : 0.8735
##              Neg Pred Value : 0.8692
##              Prevalence : 0.5611
##              Detection Rate : 0.5061
##      Detection Prevalence : 0.5794
##              Balanced Accuracy : 0.8675
##
##              'Positive' Class : neutral or dissatisfied
##

```

This visualization presents the confusion matrix as a heatmap. The high values along the diagonal indicate that most passengers were classified correctly, confirming the effectiveness of the model.

```
cm_df <- as.data.frame(cm$table)

ggplot(cm_df, aes(Prediction,
  Reference, fill =
  Freq)) +
  geom_tile() +
  geom_text(aes(label=Freq),
    size=5)
```



ROC Curve

```
roc_obj <- roc(test$satisfaction, prob)
```

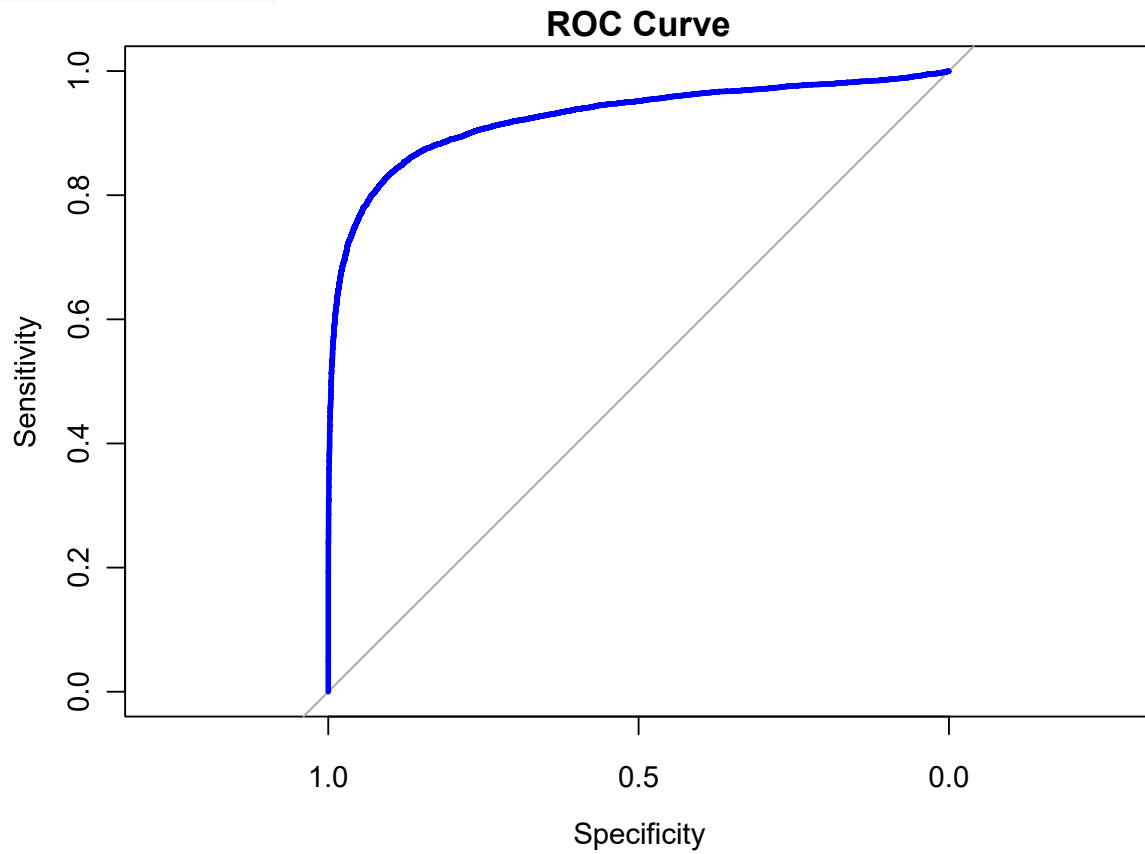
```
## Setting levels: control = neutral or dissatisfied, case = satisfied
```

```
## Setting direction: controls < cases
```

The ROC curve evaluates the model's ability to distinguish between satisfied and dissatisfied passengers across different classification thresholds. A curve closer to the top-left corner indicates better performance. The AUC value of 0.9255 indicates excellent classification performance. This

means the model can effectively differentiate between satisfied and dissatisfied passengers with high accuracy.

```
plot(
  roc_obj, col="blue", lwd=3,
  main="ROC Curve"
)
```



```
auc(roc_obj)
```

Area under the curve: 0.9255

KMO and Bartlett Test

The KMO test assesses whether the data is suitable for factor analysis. The obtained value of 0.77 indicates good sampling adequacy and confirms that factor analysis can be performed reliably. Bartlett's Test checks whether significant correlations exist among the selected variables. The significant result confirms that the variables are sufficiently related for factor analysis.

This code selects important service-related variables such as WiFi, online booking, seat comfort, entertainment, and cleanliness for factor analysis to identify underlying service quality dimensions.

The selected variables are standardized so that each feature contributes equally to factor analysis. This prevents variables with larger values from dominating the results

```
fa_vars <- train[, c(
  "Inflight.wifi.service", "Departure.Arrival.time.convenient",
  "Ease.of.Online.booking",
  "Gate.location",
  "Food.and.drink",
  "Online.boarding",
  "Seat.comfort",
  "Inflight.entertainment",
  "On.board.service",
  "Baggage.handling",
  "Inflight.service",
  "Cleanliness"
)] fa_scaled <- scale(fa_vars)
```

```
KMO(fa_scaled)
```

```
## Kaiser-Meyer-Olkin factor adequacy
```

```
## Call: KMO(r = fa_scaled)
```

```
## Overall MSA = 0.77
```

```
## MSA for each item =
```

```
##           Inflight.wifi.service Departure.Arrival.time.convenient
##                   0.73                                           0.76
##           Ease.of.Online.booking                               Gate.location
##                   0.68                                           0.70
##           Food.and.drink                                       Online.boarding
##                   0.84                                           0.72
##           Seat.comfort                                         Inflight.entertainment
```



```
##                                0.83                                0.78
##                                On.board.service                    Baggage.handling
##                                0.80                                0.77
##                                Inflight.service                    Cleanliness
##                                0.74                                0.82
```

```
cortest.bartlett(
  cor(fa_scaled), n =
  nrow(fa_scaled)
)
```

```
## $chisq
## [1] 554091.6
##
## $p.value
## [1] 0
##
## $df ##
## [1] 66
```

Factor Analysis

This step extracts four underlying factors from the service quality variables. It reduces complexity by grouping related variables into broader dimensions of airline service quality.

```
fa_model_final <- fa(
  fa_scaled, nfactors = 4,
  rotate = "oblimin"
)
```

Loading required namespace: GPArotation

The factor loadings show how strongly each variable contributes to the extracted factors. The results reveal major dimensions related to cabin experience, digital services, operational services, and airport convenience.

```
print(
  fa_model_final$loadings, cutoff =
  0.4
)
```

```
##
## Loadings:
##                                MR1    MR2    MR3    MR4
## Inflight.wifi.service          0.767
```

```

## Departure.Arrival.time.convenient 0.468 ## Ease.of.Online.booking 0.838
## Gate.location                                0.623
## Food.and.drink                                0.800
## Online.boarding                                0.665      -0.402
## Seat.comfort                                0.760
## Inflight.entertainment                        0.763
## On.board.service                                0.678
## Baggage.handling                              0.765
## Inflight.service                              0.819
## Cleanliness                                0.882
##
##              MR1   MR2   MR3   MR4
## SS loadings      2.627 1.884 1.875 0.834
## Proportion Var 0.219 0.157 0.156 0.069
## Cumulative Var 0.219 0.376 0.532 0.602

```

Factor Diagram

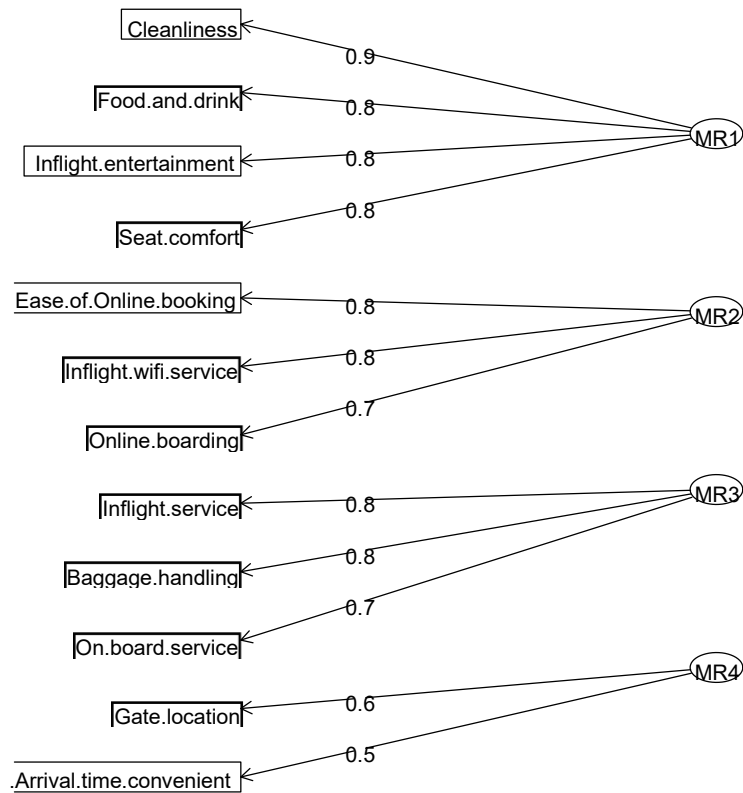
The factor diagram visually represents the relationships between service variables and the extracted factors. It provides an easy-to-understand overview of the underlying service quality structure.

```

fa.diagram(
  fa_model_final, main="Factor
Structure of Airline Service
Quality"
)

```

Factor Structure of Airline Service Quality



Online Boarding Analysis

The boxplot compares online boarding ratings across satisfaction groups. Satisfied passengers generally report higher online boarding ratings, indicating that online boarding quality plays an important role in overall passenger satisfaction.

```
ggplot(train, aes( x=satisfaction,
  y=Online.boarding,
  fill=satisfaction
)) +
  geom_boxplot() + theme_minimal()
```

